

Stock-path simulation · tables

Friday 17 March 2023 monthly expiry · 1-hour lookback · SPY / AAPL / MSFT

What this PDF is for

- This PDF records stock-path simulation results for the Friday 17 March 2023 monthly-expiry study.
- Each table compares six models on how closely the simulated path tracks the historical 1-minute price.
- Heston (no jumps) and Heston-Merton are in the tables. The matching bar-chart PDF drops both so the other four models stay readable.
- The first results page is the equal-weight mean of SPY, AAPL, and MSFT. Later pages repeat the layout for each ticker.
- In every block the best model is bold and shaded: lowest RMSE of the median (50th-percentile) path versus actual S_t .

Methodology

- Lookback is 1 hour (60 consecutive regular-trading-hours minutes) at every calibration time.
- Rolling is either hourly (parameters held for one hour) or minutely (parameters updated every minute).
- Models: GBM, Merton jump-diffusion, Heston (no jumps), Heston-Merton, GARCH(1,1), and GARCH-Merton. Same 1-minute notebooks as the rest of V2.
- Each name is simulated with 1,000 Monte Carlo paths and a common seed of 42.
- The reported path is the 50th percentile across paths at each minute — not the Monte Carlo mean.
- RMSE and MAE compare that median path with historical S_t on the continuous RTH clock (overnight and weekend gaps removed).
- ICP is the share of historical prices that fall inside the 25th–75th percentile band.
- Average band width is the mean of (75th – 25th percentile) over the window.

Stock-path metrics · mean of SPY / AAPL / MSFT

1-day · hourly rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	1.415	1.179	82.1%	4.119
Merton	1.482	1.236	82.0%	4.113
Heston	2.370	2.001	57.2%	4.220
Heston-Merton	2.322	1.949	58.9%	4.238
GARCH	1.343	1.123	86.6%	4.797
GARCH-Merton	1.348	1.128	86.6%	4.803

1-day · minutely rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	0.886	0.699	86.8%	3.953
Merton	0.909	0.730	86.2%	3.872
Heston	1.430	1.210	75.5%	4.008
Heston-Merton	1.459	1.232	76.5%	3.993
GARCH	0.944	0.746	87.0%	3.913
GARCH-Merton	0.921	0.726	87.2%	3.871

5-day · hourly rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	2.605	2.226	97.4%	10.121
Merton	1.978	1.644	92.8%	9.675
Heston	9.978	9.066	12.5%	10.673
Heston-Merton	5.028	4.356	61.9%	10.379
GARCH	2.235	1.826	94.8%	11.336
GARCH-Merton	2.320	1.915	94.6%	11.167

5-day · minutely rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	1.042	0.791	97.4%	9.750
Merton	2.036	1.761	95.1%	9.598
Heston	5.626	4.779	60.5%	10.354
Heston-Merton	5.210	4.292	78.0%	10.285
GARCH	1.379	1.128	96.1%	9.833
GARCH-Merton	1.344	1.088	96.2%	9.927

Bold shaded row = lowest RMSE (median path vs S_t). ICP = % of actual prices inside the 25th-75th band. Heston-Merton is included. Mean is equal-weight across the three names.

Stock-path metrics · SPY

1-day · hourly rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	1.270	1.073	82.1%	4.243
Merton	1.378	1.168	82.1%	4.225
Heston	1.421	1.234	81.0%	4.088
Heston-Merton	1.352	1.153	81.5%	4.107
GARCH	1.522	1.277	85.1%	5.134
GARCH-Merton	1.522	1.277	85.1%	5.134

1-day · minutely rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	0.980	0.724	81.8%	4.222
Merton	0.992	0.759	80.5%	4.138
Heston	1.085	0.868	81.0%	4.131
Heston-Merton	1.025	0.826	82.1%	4.016
GARCH	0.987	0.728	80.8%	4.161
GARCH-Merton	0.980	0.728	81.0%	4.072

5-day · hourly rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	3.245	2.780	98.5%	12.609
Merton	1.796	1.463	94.6%	12.325
Heston	12.918	12.144	3.9%	13.315
Heston-Merton	6.361	5.881	68.1%	12.838
GARCH	2.240	1.725	93.2%	13.122
GARCH-Merton	2.425	1.923	93.5%	13.168

5-day · minutely rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	1.217	0.938	97.8%	12.250
Merton	3.146	2.790	94.9%	11.997
Heston	2.681	2.256	94.2%	12.657
Heston-Merton	4.411	3.644	90.8%	12.420
GARCH	1.597	1.296	95.9%	12.575
GARCH-Merton	1.554	1.244	96.1%	12.389

Bold shaded row = lowest RMSE (median path vs S_t). ICP = % of actual prices inside the 25th-75th band. Heston-Merton is included.

Stock-path metrics · AAPL

1-day · hourly rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	0.621	0.511	94.9%	2.465
Merton	0.712	0.587	94.6%	2.465
Heston	1.608	1.416	47.2%	2.564
Heston-Merton	1.532	1.340	51.8%	2.597
GARCH	0.636	0.532	88.2%	2.250
GARCH-Merton	0.653	0.548	88.2%	2.268

1-day · minutely rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	0.418	0.346	94.1%	2.402
Merton	0.413	0.345	94.1%	2.325
Heston	0.566	0.448	87.9%	2.409
Heston-Merton	0.558	0.426	87.9%	2.416
GARCH	0.400	0.330	94.4%	2.388
GARCH-Merton	0.398	0.333	94.4%	2.336

5-day · hourly rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	2.096	1.892	96.8%	6.274
Merton	1.288	1.117	97.5%	5.930
Heston	5.662	5.236	14.2%	6.550
Heston-Merton	3.488	3.011	45.3%	6.441
GARCH	1.525	1.317	98.8%	6.703
GARCH-Merton	1.366	1.170	98.9%	6.617

5-day · minutely rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	0.506	0.396	98.3%	5.991
Merton	0.632	0.508	96.8%	6.005
Heston	2.303	1.961	71.7%	6.305
Heston-Merton	2.071	1.693	95.3%	6.367
GARCH	0.536	0.415	98.2%	5.947
GARCH-Merton	0.546	0.432	98.3%	6.070

Bold shaded row = lowest RMSE (median path vs S_t). ICP = % of actual prices inside the 25th-75th band. Heston-Merton is included.

Stock-path metrics · MSFT

1-day · hourly rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	2.355	1.953	69.2%	5.650
Merton	2.355	1.953	69.2%	5.650
Heston	4.081	3.354	43.3%	6.008
Heston-Merton	4.081	3.354	43.3%	6.008
GARCH	1.871	1.559	86.4%	7.007
GARCH-Merton	1.871	1.559	86.4%	7.007

1-day · minutely rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	1.262	1.027	84.4%	5.236
Merton	1.321	1.085	84.1%	5.154
Heston	2.638	2.314	57.4%	5.483
Heston-Merton	2.794	2.445	59.5%	5.547
GARCH	1.446	1.181	85.9%	5.191
GARCH-Merton	1.384	1.117	86.2%	5.203

5-day · hourly rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	2.475	2.005	96.9%	11.480
Merton	2.850	2.352	86.3%	10.769
Heston	11.354	9.817	19.4%	12.154
Heston-Merton	5.234	4.177	72.4%	11.859
GARCH	2.940	2.436	92.3%	14.185
GARCH-Merton	3.169	2.651	91.3%	13.716

5-day · minutely rolling

Model	RMSE	MAE	ICP	Avg band width
GBM	1.402	1.041	96.0%	11.009
Merton	2.331	1.986	93.4%	10.791
Heston	11.893	10.121	15.7%	12.101
Heston-Merton	9.148	7.539	47.8%	12.069
GARCH	2.005	1.671	94.2%	10.978
GARCH-Merton	1.931	1.589	94.3%	11.321

Bold shaded row = lowest RMSE (median path vs S_t). ICP = % of actual prices inside the 25th-75th band. Heston-Merton is included.